Inference in OLS

Econometrics. Lecture 3

Content

1. Sampling distributions of the OLS estimators
2. Testing hypotheses about a single population parameter: the t-test
3. Confidence intervals
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1. Sampling distributions of the OLS estimators

Assumption MLR.6. Normality

In order to obtain tractable distributions of model parameters we assume normality of the residuals.

Assumption MLR.6

The population error u is independent of the explanatory variables $x_1, x_2, \ldots, x_k$ and is normally distributed with zero mean and variance σ^2: $u \sim \text{Normal}(0, \sigma^2)$.

In cross-sectional settings assumptions MLR.1 through MLR.6 are called the **classical linear model (CLM) assumptions**.

The OLS estimators are the **minimum variance unbiased estimators**, which means that OLS has the smallest variance among all (and not just linear in the y_i) unbiased estimators.

Normal sampling distributions

Under the CLM assumptions MLR.1 through MLR.6, conditional on the sample values of the independent variables,

$$\hat{\beta}_j \sim \text{Normal}[\beta_j, \text{Var}(\hat{\beta}_j)]$$

Therefore $\dfrac{\hat{\beta}_j - \beta_j}{sd(\hat{\beta}_j)} \sim \text{Normal}(0,1)$

And $\dfrac{\hat{\beta}_j - \beta_j}{se(\hat{\beta}_j)} \sim t_{n-k-1} = t_{df}$

where k is the number of slope parameters, and $n - k - 1$ is the degrees of freedom (df).

2. Testing hypotheses about a single population parameter: the t-test

Testing against one-sided alternatives. The null hypothesis

In many applications we are interested in testing the null hypothesis

$$\mathrm{H}_0\colon \beta_j = 0,$$ where *j* corresponds to any of the *k* independent variables.

This implies that, once all other $x_1, x_2, \ldots, x_{j-1}, x_{j+1}, \ldots, x_k$ have been accounted for, x_j has *no effect* on the expected value of *y*.

The null hypothesis $\mathrm{H}_0\colon \beta_2 = 0$ means that, once education and tenure have been accounted for, the number of years in the workforce has no effect on hourly wage.

$$\log(wage) = \beta_0 + \beta_1 educ + \beta_2 exper + \beta_3 tenure + u$$

If, on the other hand, $\beta_2 > 0$, then prior work experience contributes to productivity, and hence to wage

Testing against one-sided alternatives. The alternative hypothesis

Consider a one-sided alternative of the form

$$\mathrm{H}_1\colon \beta_j > 0 \qquad \text{or} \qquad \mathrm{H}_1\colon \beta_j < 0$$

Rejection rule (one-tailed test)

1. Decide on the **significance level** (α). 5% significance level ($\alpha = 0.05$) is the most popular.
2. Calculate the ***t*-statistic** ($t_{\hat{\beta}_j}$) by $(\hat{\beta}_j - \beta_j)/se(\hat{\beta}_j)$, where because of $\beta_j = 0$ under null it becomes $\hat{\beta}_j/se(\hat{\beta}_j)$.
3. Find from *t*-tables the *t*-critical (c) for $n - k - 1$ degrees of freedom.
4. For $\mathrm{H}_1\colon \beta_j > 0$ reject the null if $t_{\hat{\beta}_j} > c$. For $\mathrm{H}_1\colon \beta_j < 0$ reject the null if $t_{\hat{\beta}_j} < -c$.

Testing against one-sided alternatives. Wage example

$$\widehat{\log(wage)} = 0.284 + 0.092educ + 0.004exper + 0.022tenure$$

Dependent Variable: LOG(WAGE)
Method: Least Squares
Date: 12/30/25 Time: 14:35
Sample: 1 526
Included observations: 526

Variable	Coefficient	Std. Error	t-Statistic	Prob.
EDUC	0.092029	0.007330	12.55525	0.0000
EXPER	0.004121	0.001723	2.391437	0.0171
TENURE	0.022067	0.003094	7.133070	0.0000
C	0.284360	0.104190	2.729230	0.0066

R-squared	0.316013	Mean dependent var	1.623268
Adjusted R-squared	0.312082	S.D. dependent var	0.531538
S.E. of regression	0.440862	Akaike info criterion	1.207406
Sum squared resid	101.4556	Schwarz criterion	1.239842
Log likelihood	-313.5478	Hannan-Quinn criter.	1.220106
F-statistic	80.39092	Durbin-Watson stat	1.768805
Prob(F-statistic)	0.000000		

$H_0: \beta_{exper} = 0 \quad H_1: \beta_{exper} > 0$

$t_{\hat{\beta}_j} = 2.39 \quad > \quad c = 1.648$

Reject H_0. β_{exper} is statistically greater than zero at 5% significance level.

To find critical value for a t-stat in EViews type "@qtdist(1- α, df)". In our case "@qtdist(0.95, 522)".

Testing against two-sided alternatives

In most applications, it is common to test the null hypothesis against a **two-sided alternative**.

$$\mathrm{H}_0\colon \beta_j = 0 \qquad \mathrm{H}_1\colon \beta_j \neq 0$$

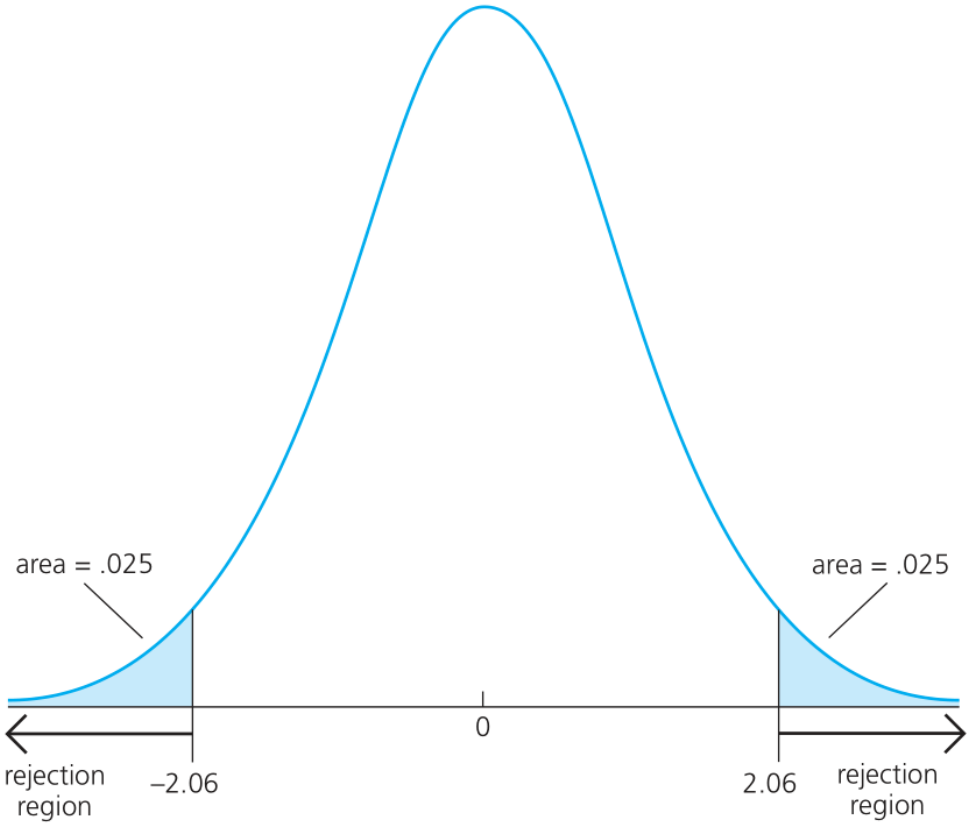
The **rejection rule** for $\mathrm{H}_0\colon \beta_j = 0$ against $\mathrm{H}_1\colon \beta_j \neq 0$ is

$$|t_{\hat{\beta}_j}| > c$$

If H_0 is rejected in favor of H_1 at the 5% level, we usually say that "β_j is **statistically significant**, or statistically different from zero, at the 5% level."

If H_0 is not rejected, we say that "β_j is **statistically insignificant** at the 5% level."

5% rejection rule for the alternative $H_1\colon \beta_j \neq 0$ with 25 df



Testing against two-sided alternatives. Wage example

After running the wage equation

Dependent Variable: LOG(WAGE)
Method: Least Squares
Date: 12/30/25 Time: 14:35
Sample: 1 526
Included observations: 526

Variable	Coefficient	Std. Error	t-Statistic	Prob.
EDUC	0.092029	0.007330	12.55525	0.0000
EXPER	0.004121	0.001723	2.391437	0.0171
TENURE	0.022067	0.003094	7.133070	0.0000
C	0.284360	0.104190	2.729230	0.0066

R-squared	0.316013	Mean dependent var	1.623268
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Sum squared resid	101.4556	Schwarz criterion	1.239842
Log likelihood	-313.5478	Hannan-Quinn criter.	1.220106
F-statistic	80.39092	Durbin-Watson stat	1.768805
Prob(F-statistic)	0.000000		

Question: Which coefficients are significant at 1% level if the critical value with 522 degrees of freedom is 2.585?

A rule of thumb of significance tests

For a two-tail test the critical value for a 5% level of significance and for 30 degrees of freedom is 2.042, while for 60 df is 2.00.

Therefore, for large samples it is quite safe to use as a **rule of thumb** a critical value of $|t| > 2$.

Coefficient is significant at 5% level if *t*-statistic is greater than 2 in absolute value.

Testing other hypothesis

If the null is stated as $\mathrm{H}_0\colon \beta_j = a_j$, where a_j is hypothesized value of β_j, then

$$t = (\hat{\beta}_j - a_j)/se(\hat{\beta}_j).$$

The **general *t*-statistic** is usefully written as

$$t = \frac{(estimate - hypothesized\ value)}{standard\ error}$$

We can use the general t-statistic to test against one-sided or two-sided alternatives.

Testing other hypothesis. Campus crime and enrollment (1)

After running the equation $\mathrm{H}_0: \beta_1 = 1$ $\mathrm{H}_1: \beta_1 > 1$

Dependent Variable: LOG(CRIME)
Method: Least Squares
Date: 12/30/25 Time: 19:38
Sample: 1 97
Included observations: 97

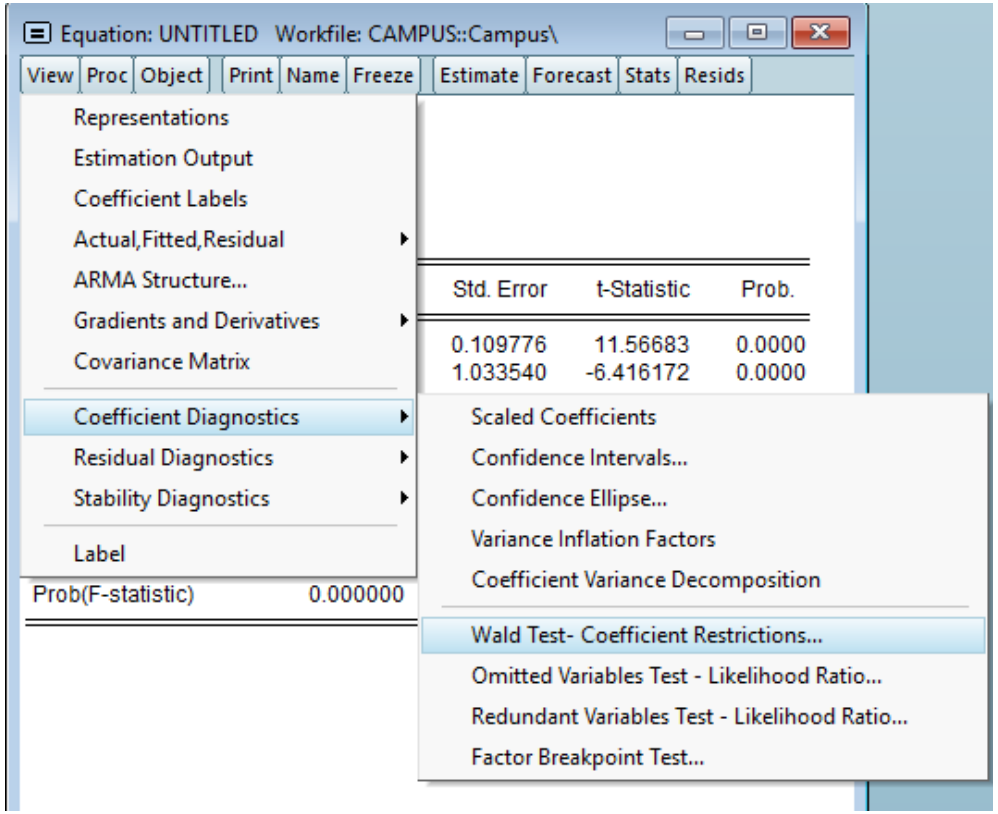
Variable	Coefficient	Std. Error	t-Statistic	Prob.
LOG(ENROLL)	1.269760	0.109776	11.56683	0.0000
C	-6.631370	1.033540	-6.416172	0.0000

R-squared	0.584775	Mean dependent var	5.277149
Adjusted R-squared	0.580404	S.D. dependent var	1.381121
S.E. of regression	0.894638	Akaike info criterion	2.635608
Sum squared resid	76.03581	Schwarz criterion	2.688694
Log likelihood	-125.8270	Hannan-Quinn criter.	2.657073
F-statistic	133.7915	Durbin-Watson stat	1.548138
Prob(F-statistic)	0.000000		

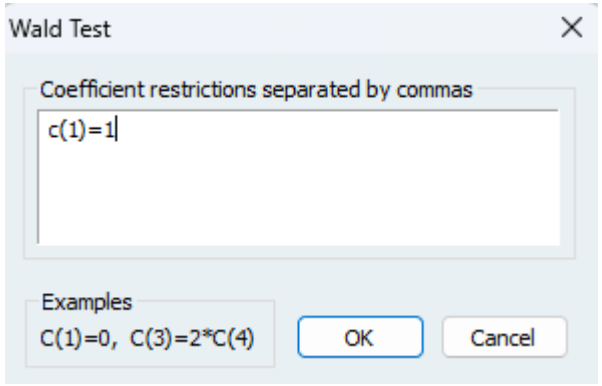
We test whether the elasticity of crime with respect to enrollment is one. This means that a 1% increase in enrollment leads to, on average, a 1% increase in crime.

Testing other hypothesis. Campus crime and enrollment (2)

Use the Wald test



Specify H_0



Analyze results

Wald Test:
Equation: Untitled

Test Statistic	Value	df	Probability
t-statistic	2.457371	95	0.0158
F-statistic	6.038671	(1, 95)	0.0158
Chi-square	6.038671	1	0.0140

Null Hypothesis: C(1)=1
Null Hypothesis Summary:

Normalized Restriction (= 0)	Value	Std. Err.
-1 + C(1)	0.269760	0.109776

Restrictions are linear in coefficients.

$$t_{\hat{\beta}_1} = 2.46 \quad > \quad c = 1.66$$

Reject H_0: $\beta_1 = 1$ in favor of H_1: $\beta_1 > 1$ at the 5% level

The notion of *p*-value

The ***p*-value** shows what is the smallest significance level at which the null hypothesis would be rejected.

For example, if the *p*-value is 0.50, then we would observe a value of the *t*-statistic as extreme as we did in 50% of all random samples when the null hypothesis is true; this is pretty weak evidence against H_0.

Dependent Variable: LOG(WAGE)
Method: Least Squares
Date: 12/30/25 Time: 14:35
Sample: 1 526
Included observations: 526

Question: Which coefficients are significant at 1% level?

Variable	Coefficient	Std. Error	t-Statistic	Prob.
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EXPER	0.004121	0.001723	2.391437	0.0171
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Prob(F-statistic)	0.000000		

Once the *p*-value has been computed, a classical test can be carried out at any desired level. If α denotes the significance level of the test (in decimal form), then **H_0 is rejected if *p*-value < α**; otherwise, H_0 is not rejected at the $100 * \alpha$ level.

Statistical vs economic significance

We should pay attention to the magnitude (and sign) of the coefficient estimates in addition to the size of the *t*-statistics.

The **statistical significance** of a variable x_j is determined entirely by the size of *t*-statistic.

The **economic significance** (or practical significance) of a variable is related to the size (and sign) of coefficient β_j.

With large sample sizes, standard errors are often quite small relative to the coefficient estimates resulting in statistical significance.

Guidelines for discussing the economic and statistical significance

1. Check for statistical significance. If the variable is statistically significant, discuss the magnitude (and sign) of the coefficient to get an idea of its practical or economic importance.
2. If a variable is not statistically significant at the usual levels (10%, 5%, or 1%), you might still ask if the variable has the expected effect on *y* and whether that effect is practically large. If it is large, you should compute a *p*-value for the *t*-statistic. For small sample sizes, you can sometimes make a case for *p*-values as large as 0.20 (but there are no hard rules).
3. A significant variable that has the unexpected sign and a practically large effect is troubling and difficult to resolve. One must usually think more about the model and the nature of the data to solve such problems. Often, a counterintuitive, significant estimate results from the omission of a key variable.

3. Confidence intervals

Confidence intervals

Confidence Intervals (CI) – provide a range of likely values for the population parameter, and not just a point estimate.

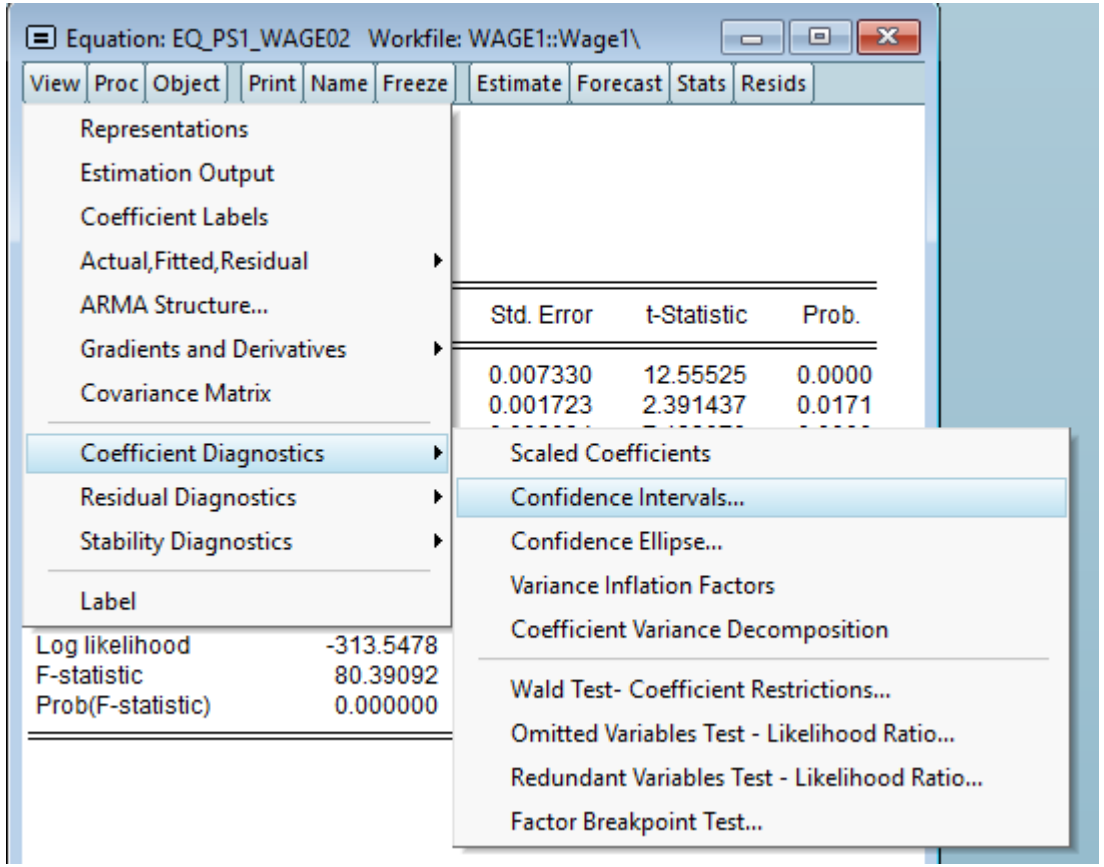
A **95% confidence interval:** $\hat{\beta}_j \pm c \cdot \text{se}(\hat{\beta}_j)$

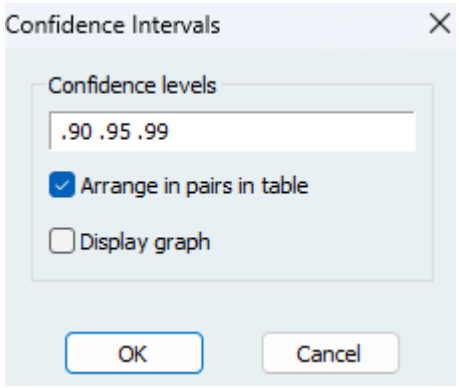
c is the 97.5th percentile in a t_{n-k-1} distribution

When $n - k - 1 > 50$, the value of *c* is so close to 2 that we can use a simple rule of thumb for a 95% confidence interval: $\hat{\beta}_j$ plus or minus 2 of its standard errors.

Once a confidence interval is constructed, it is easy to carry out two-tailed hypotheses tests. If the null hypothesis is H_0: $\beta_j = a_j$, then H_0 is rejected against H_1: $\beta_j \neq a_j$ at (say) the 5% significance level if, and only if, a_j is not in the 95% confidence interval.

Confidence intervals.
EViews example: Wage equation





Coefficient Confidence Intervals
Date: 12/30/25 Time: 22:17
Sample: 1 526
Included observations: 526

		90% CI		95% CI		99% CI	
Variable	Coefficient	Low	High	Low	High	Low	High
EDUC	0.092029	0.079951	0.104107	0.077629	0.106429	0.073079	0.110979
EXPER	0.004121	0.001282	0.006961	0.000736	0.007507	-0.000334	0.008576
TENURE	0.022067	0.016970	0.027165	0.015990	0.028145	0.014069	0.030065
C	0.284360	0.112677	0.456042	0.079676	0.489044	0.014998	0.553721

4. Testing hypotheses about a single linear combination of the parameters

The equality of two coefficients

$$\log(wage) = \beta_0 + \beta_1 jc + \beta_2 univ + \beta_3 exper + u$$

where jc = number of years attending a two-year college; $univ$ = number of years at a four-year college; $exper$ = months in the workforce.

$\mathrm{H}_0: \beta_1 = \beta_2$ $\quad$ $\mathrm{H}_1: \beta_1 < \beta_2$

We can rewrite the null as $\mathrm{H}_0: \beta_1 - \beta_2 = 0$ and alternative as $\mathrm{H}_1: \beta_1 - \beta_2 < 0$,

The *t*-statistics $t = \dfrac{\hat{\beta}_1 - \hat{\beta}_2}{\mathrm{se}(\hat{\beta}_1 - \hat{\beta}_2)}.$

For this hypothesis one can compute the standard error as

$$\mathrm{se}(\hat{\beta}_1 - \hat{\beta}_2) = \left\{ [\mathrm{se}(\hat{\beta}_1)]^2 + [\mathrm{se}(\hat{\beta}_2)]^2 - 2s_{12} \right\}^{1/2}$$

where s_{12} denotes an estimate of $\mathrm{Cov}(\hat{\beta}_1, \hat{\beta}_2)$.

The rejection rule is of the form $t < -c$

where *c* is a positive value chosen from the appropriate *t*-distribution.

The equality of two coefficients. EViews example

After running the equation

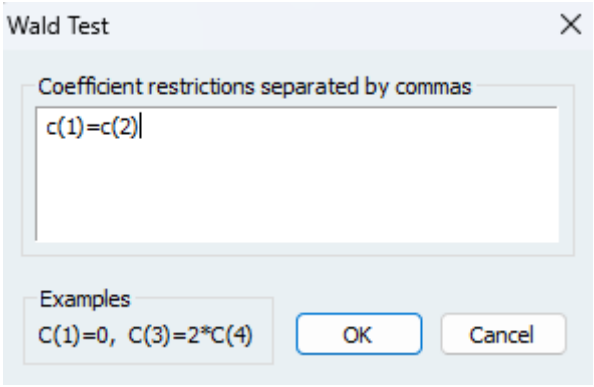
Dependent Variable: LWAGE
Method: Least Squares
Date: 12/30/25 Time: 22:29
Sample: 1 6763
Included observations: 6763

Variable	Coefficient	Std. Error	t-Statistic	Prob.
JC	0.066697	0.006829	9.766984	0.0000
UNIV	0.076876	0.002309	33.29808	0.0000
EXPER	0.004944	0.000157	31.39717	0.0000
C	1.472326	0.021060	69.91020	0.0000

R-squared	0.222442	Mean dependent var	2.248096
Adjusted R-squared	0.222097	S.D. dependent var	0.487692
S.E. of regression	0.430138	Akaike info criterion	1.151172
Sum squared resid	1250.544	Schwarz criterion	1.155205
Log likelihood	-3888.687	Hannan-Quinn criter.	1.152564
F-statistic	644.5330	Durbin-Watson stat	1.968444
Prob(F-statistic)	0.000000		

$\hat{\beta}_1 - \hat{\beta}_2 = -0.0102$

Run the Wald test



Specify H_0: $\beta_1 = \beta_2$.

Analyze *t*-statistic

p-value for a two-sided test! Divide by 2 to get *p*-value for a one-sided test

Wald Test:
Equation: Untitled

Test Statistic	Value	df	Probability
t-statistic	-1.467657	6759	0.1422
F-statistic	2.154016	(1, 6759)	0.1422
Chi-square	2.154016	1	0.1422

Null Hypothesis: C(1)=C(2)
Null Hypothesis Summary:

Normalized Restriction (= 0)	Value	Std. Err.
C(1) - C(2)	-0.010180	0.006936

Restrictions are linear in coefficients.

$t = -1.468 > -c = -1.645$

p-value = 0.07 > 0.05

We fail to reject H_0: $\beta_1 = \beta_2$.

5. Testing multiple linear restrictions: the F-test

Joint hypotheses test. Hypothesis

Major league baseball players' salaries model

$$\log(salary) = \beta_0 + \beta_1 years + \beta_2 gamesyr + \beta_3 bavg + \beta_4 hrunsyr + \beta_5 rbisyr + u$$

where *salary* is the 1993 total salary, *years* is years in the league, *gamesyr* is average games played per year, *bavg* is career batting average (for example, *bavg* =250), *hrunsyr* is home runs per year, and *rbisyr* is runs batted in per year.

Suppose we want to test the null hypothesis that, once years in the league and games per year have been controlled for, the statistics measuring performance – *bavg*, *hrunsyr*, and *rbisyr* – have no effect on salary.

$$\mathrm{H}_0: \beta_3 = 0, \beta_4 = 0, \beta_5 = 0.$$

A test of multiple restrictions is called a **multiple hypotheses test** or a **joint hypotheses test**.

If what we have in mind is that "performance statistics matter, even after controlling for years in the league and games per year," then the appropriate alternative is simply

$$\mathrm{H}_1: \mathrm{H}_0 \text{ is not true}$$

Joint hypotheses test. *F*-statistics

The **unrestricted model**

$$\log(salary) = \beta_0 + \beta_1 years + \beta_2 gamesyr + \beta_3 bavg + \beta_4 hrunsyr + \beta_5 rbisyr + u$$

The **restricted model**

$$\log(salary) = \beta_0 + \beta_1 years + \beta_2 gamesyr + u$$

The ***F*-statistic** (or *F*-ratio) is defined by $$F \equiv \frac{(\mathrm{SSR}_r - \mathrm{SSR}_{ur})/q}{\mathrm{SSR}_{ur}/(n-k-1)},$$

where SSR_r is the sum of squared residuals from the restricted model and SSR_{ur} is the sum of squared residuals from the unrestricted model. q is the number of restrictions imposed in moving from the unrestricted to the restricted model (q independent variables are dropped).

Joint hypotheses test. Rejection rule

Under H_0 (and assuming the CLM assumptions hold), *F* is distributed as an *F* random variable with $(q, n-k-1)$ degrees of freedom.

$$F \sim F_{q,n-k-1}$$

The rejection rule is: once *c* (critical value) has been obtained, we reject H_0 in favor of H_1 at the chosen significance level if

$$F > \mathrm{c}$$

If H_0 is rejected, then we say that variables are **jointly statistically** significant at the appropriate significance level.

If H_0 is not rejected, then the variables are **jointly insignificant**, which often justifies dropping them from the model.

Joint hypotheses test. EViews example (1)

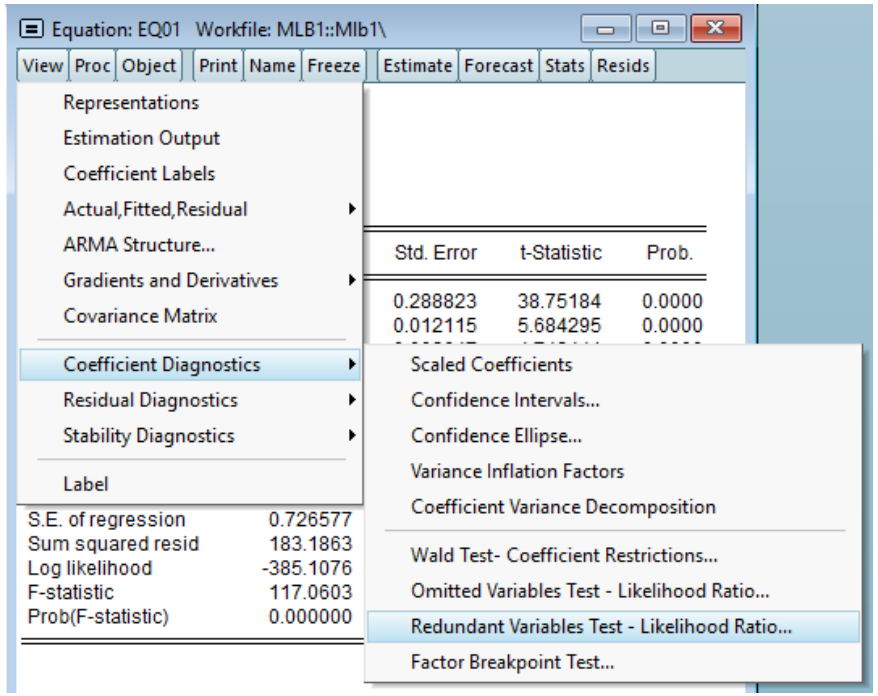
Unrestricted model

Dependent Variable: LOG(SALARY)
Method: Least Squares
Date: 12/31/25 Time: 13:40
Sample: 1 353
Included observations: 353

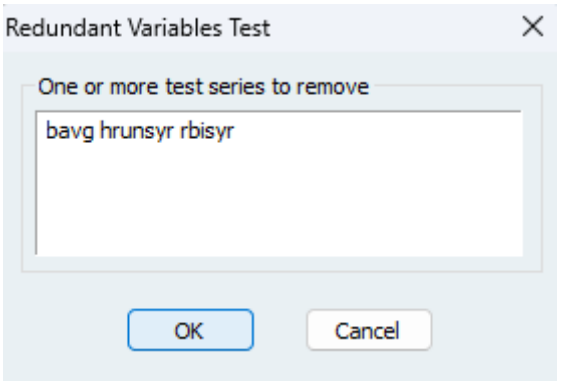
Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	11.19242	0.288823	38.75184	0.0000
YEARS	0.068863	0.012115	5.684295	0.0000
GAMESYR	0.012552	0.002647	4.742441	0.0000
BAVG	0.000979	0.001104	0.886811	0.3758
HRUNSYR	0.014429	0.016057	0.898642	0.3695
RBISYR	0.010766	0.007175	1.500459	0.1344

R-squared	0.627803	Mean dependent var	13.49218
Adjusted R-squared	0.622440	S.D. dependent var	1.182466
S.E. of regression	0.726577	Akaike info criterion	2.215907
Sum squared resid	183.1863	Schwarz criterion	2.281626
Log likelihood	-385.1076	Hannan-Quinn criter.	2.242057
F-statistic	117.0603	Durbin-Watson stat	1.265390
Prob(F-statistic)	0.000000		

Joint *F*-test



Null hypothesis



Alternatively, you can use Wald test

Joint hypotheses test. EViews example (2)

$F = 9.55 \quad > \quad c = 2.63$

p-value = 0.00 $<$ 0.05

We reject H_0: $\beta_3 = 0$, $\beta_4 = 0$, $\beta_5 = 0$.

At least one of the performance statistics measure matters for salary after controlling for years in the league and games per year

(to find critical value for a F-stat in EViews type "@qfdist(1- α, q, n-k-1)". In our case "@qfdist(0.95, 3, 347)".

Results of the test

Redundant Variables Test
Equation: EQ01
Redundant variables: BAVG HRUNSYR RBISYR
Specification: LOG(SALARY) C YEARS GAMESYR BAVG HRUNSYR RBISYR
Null hypothesis: BAVG HRUNSYR RBISYR are jointly insignificant

	Value	df	Probability
F-statistic	9.550254	(3, 347)	0.0000
Likelihood ratio	28.00529	3	0.0000

F-test summary:

	Sum of Sq.	df	Mean Squares
Test SSR	15.12515	3	5.041717
Restricted SSR	198.3115	350	0.566604
Unrestricted SSR	183.1863	347	0.527914

LR test summary:

	Value
Restricted LogL	-399.1103
Unrestricted LogL	-385.1076

Restricted Test Equation:
Dependent Variable: LOG(SALARY)
Method: Least Squares
Date: 12/31/25 Time: 13:43
Sample: 1 353
Included observations: 353

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	11.22380	0.108312	103.6247	0.0000
YEARS	0.071318	0.012505	5.703153	0.0000
GAMESYR	0.020174	0.001343	15.02341	0.0000

R-squared	0.597072	Mean dependent var	13.49218
Adjusted R-squared	0.594769	S.D. dependent var	1.182466
S.E. of regression	0.752731	Akaike info criterion	2.278245
Sum squared resid	198.3115	Schwarz criterion	2.311105
Log likelihood	-399.1103	Hannan-Quinn criter.	2.291320
F-statistic	259.3204	Durbin-Watson stat	1.193944
Prob(F-statistic)	0.000000		

F-statistic for overall significance

A common pessimistic hypothesis when *none* of the explanatory variables has an effect on *y*.

$$H_0: \beta_1 = \beta_2 = \ldots = \beta_k = 0,$$

and the alternative is that at least one of the β_j is different from zero.

The *F*-statistic for testing the null can be written as

$$\frac{R^2/k}{(1-R^2)/(n-k-1)},$$

where R^2 is just the usual R-squared from the regression of *y* on $x_1, x_2, \ldots, x_k$.

If we fail to reject the null, then there is no evidence that any of the independent variables help to explain *y*.

F-statistic for overall significance. EViews example

Major league baseball players' salaries model

Dependent Variable: LOG(SALARY)
Method: Least Squares
Date: 12/31/25 Time: 13:40
Sample: 1 353
Included observations: 353

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	11.19242	0.288823	38.75184	0.0000
YEARS	0.068863	0.012115	5.684295	0.0000
GAMESYR	0.012552	0.002647	4.742441	0.0000
BAVG	0.000979	0.001104	0.886811	0.3758
HRUNSYR	0.014429	0.016057	0.898642	0.3695
RBISYR	0.010766	0.007175	1.500459	0.1344

R-squared	0.627803	Mean dependent var	13.49218
Adjusted R-squared	0.622440	S.D. dependent var	1.182466
S.E. of regression	0.726577	Akaike info criterion	2.215907
Sum squared resid	183.1863	Schwarz criterion	2.281626
Log likelihood	-385.1076	Hannan-Quinn criter.	2.242057
F-statistic	117.0603	Durbin-Watson stat	1.265390
Prob(F-statistic)	0.000000		

p-value = 0.00 $<$ 0.05

We reject H_0

Testing general linear restrictions

Sometimes the restrictions implied by a theory are more complicated than just excluding some independent variables. It is still straightforward to use the *F*-statistic for testing.

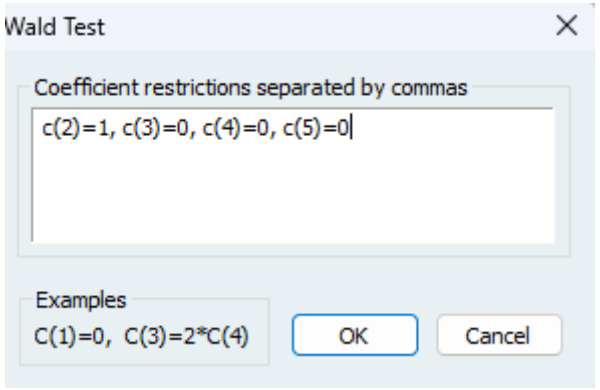
House price model

Dependent Variable: LOG(PRICE)
Method: Least Squares
Date: 12/31/25 Time: 14:59
Sample: 1 88
Included observations: 88

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.263743	0.569665	0.462980	0.6446
LOG(ASSESS)	1.043065	0.151446	6.887372	0.0000
LOG(LOTSIZE)	0.007438	0.038561	0.192884	0.8475
LOG(SQRFT)	-0.103238	0.138430	-0.745778	0.4579
BDRMS	0.033839	0.022098	1.531303	0.1295

R-squared	0.772809	Mean dependent var	5.633180
Adjusted R-squared	0.761860	S.D. dependent var	0.303573
S.E. of regression	0.148142	Akaike info criterion	-0.926147
Sum squared resid	1.821529	Schwarz criterion	-0.785389
Log likelihood	45.75048	Hannan-Quinn criter.	-0.869439
F-statistic	70.58278	Durbin-Watson stat	2.048495
Prob(F-statistic)	0.000000		

Wald test



Wald Test:
Equation: Untitled

Test Statistic	Value	df	Probability
F-statistic	0.667773	(4, 83)	0.6162
Chi-square	2.671090	4	0.6143

Null Hypothesis: C(2)=1, C(3)=0, C(4)=0, C(5)=0
Null Hypothesis Summary:

Normalized Restriction (= 0)	Value	Std. Err.
-1 + C(2)	0.043065	0.151446
C(3)	0.007438	0.038561
C(4)	-0.103238	0.138430
C(5)	0.033839	0.022098

Restrictions are linear in coefficients.

price = house price; *assess* = the assessed housing value (before the house was sold); *lotsize* = size of the lot, in feet; *sqrft* = square footage; *bdrms* = number of bedrooms.

6. Reporting regression results

Reporting regression results

1. Report the estimated OLS coefficients. The economic or practical importance of the estimates of the key variables should be discussed.
2. The standard errors should always be included along with the estimated coefficients. *(Some authors prefer to report the t-statistics or p-values)*.
3. The R-squared from the regression should always be included.
4. The number of observations used in estimating any equation should appear near the estimated equation.
5. If only a couple of models are being estimated, the results can be summarized in equation form.
6. For several models it is better to summarize the results in one or more tables.

Reporting regression results. Equation form example

House price model

$$\widehat{\log(price)} = .264 + 1.043\log(assess) + .0074\log(lotsize)$$
$$(.570)\quad (.151)\qquad\qquad (.0386)$$
$$-.1032\log(sqrft) + .0338\,bdrms$$
$$(.1384)\qquad\qquad (.0221)$$
$$n = 88,\ \text{SSR} = 1.822,\ R^2 = .773.$$

Where

price = house price.

assess = the assessed housing value (before the house was sold).

lotsize = size of the lot, in feet.

sqrft = square footage.

bdrms = number of bedrooms.

Reporting regression results. Table form example

Money growth and Inflation: Cross-section estimation

Dependent variable: Inflation (y/y)

	1990-2002	2002-2012
Constant	2.521	5.27***
	(0.77)	(4.47)
Income growth (y/y)	-0.574	-0.46**
	(-1.25)	(-2.23)
Money growth (y/y)	0.638***	0.286***
	(3.12)	(4.15)
Money growth *High inf.	0.185**	0.141***
	(2.01)	(4.46)
Money growth *High fin. dev.	0.076	-0.075**
	(0.68)	(-2.21)
Number of countries	62	63
Adjusted R^2	0.635	0.635

Sources: World Economic Outlook and IMF staff calculations.

1/ Values of t-statistics reported in the parentheses are calculated using heteroscedasticity-consistent standard errors; * significant at 10 percent; ** significant at 5 percent; *** significant at 1 percent.

Source: Lane, C. et al. (2014). *Conditionality in evolving monetary policy regimes*. IMF Policy Paper.

Summary

1. Under the classical linear model assumptions MLR.1 through MLR.6, the OLS estimators are normally distributed.

2. We use *t*-statistics to test hypotheses about a single parameter against one- or two-sided alternatives, using one- or two-tailed tests, respectively. The most common null hypothesis is H_0: $\beta_j = 0$, but we sometimes want to test other values of β_j under H_0.

3. We first choose a significance level, which, along with the *df* and alternative hypothesis, determines the critical value against which we compare the *t*-statistic. It is more informative to compute the *p*-value for a *t*-test – the smallest significance level for which the null hypothesis is rejected – so that the hypothesis can be tested at any significance level.

4. Under the CLM assumptions, confidence intervals can be constructed for each β_j.

5. The *F*-statistic is used to test multiple exclusion restrictions. The alternative for *F*-testing is two-sided. The null hypothesis is rejected when the *F*-statistic exceeds the critical value, *c*. Alternatively, we can compute a *p*-value to summarize the evidence against H_0.

Appendix.
Asymptotic normality and large sample inference

If the errors $u_1, u_2, \ldots, u_n$ don't have normal distribution, the $\hat{\beta}_j$ will not be normally distributed, which means that the *t*-statistics will not have *t*-distributions and the *F*-statistics will not have *F*-distributions.

However, we can use the central limit theorem to conclude that the OLS estimators satisfy **asymptotic normality**, which means they are approximately normally distributed in large enough sample sizes.

This is useful because the normality assumption MLR.6 has been dropped; the only restriction on the distribution of the error is that it has finite variance (errors are not "wildly explosive"), something we will always assume. We have also assumed zero conditional mean (MLR.4) and homoskedasticity of *u* (MLR.5).

Unfortunately, there are no general prescriptions on how big the sample size must be before the approximation is good enough. Some econometricians think that ***n* = 30** is satisfactory, but this cannot be sufficient for all possible distributions of *u*.

Consistency

The Nobel Prize–winning econometrician Clive W. J. Granger once remarked, *"If you can't get it right as n goes to infinity, you shouldn't be in this business."*

Consistency involves a thought experiment about what would happen as the sample size gets large (while, at the same time, we obtain numerous random samples for each sample size). If obtaining more and more data does not generally get us closer to the parameter value of interest, then we are using a poor estimation procedure.

Sampling distributions of $\widehat{\boldsymbol{\beta}}_1$ for sample sizes $\boldsymbol{n_1 < n_2 < n_3}$.

